7/3/2026

Contrastive learning

A Simple framework for Contrastive Learning of Visual Representations ← (P)

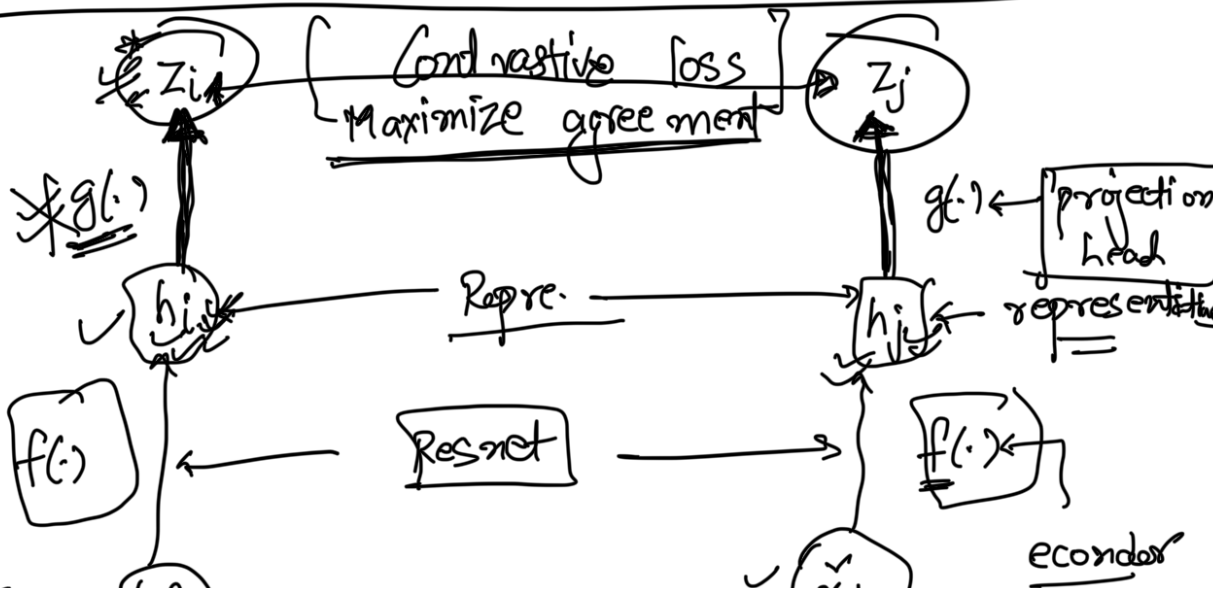
Contrastive learning : It is a method that focuses on extracting meaningful representations by contrasting positive and negative pairs of instances.

P → 1. Data **Augmentations** : Composition of data augmentation plays an important role.

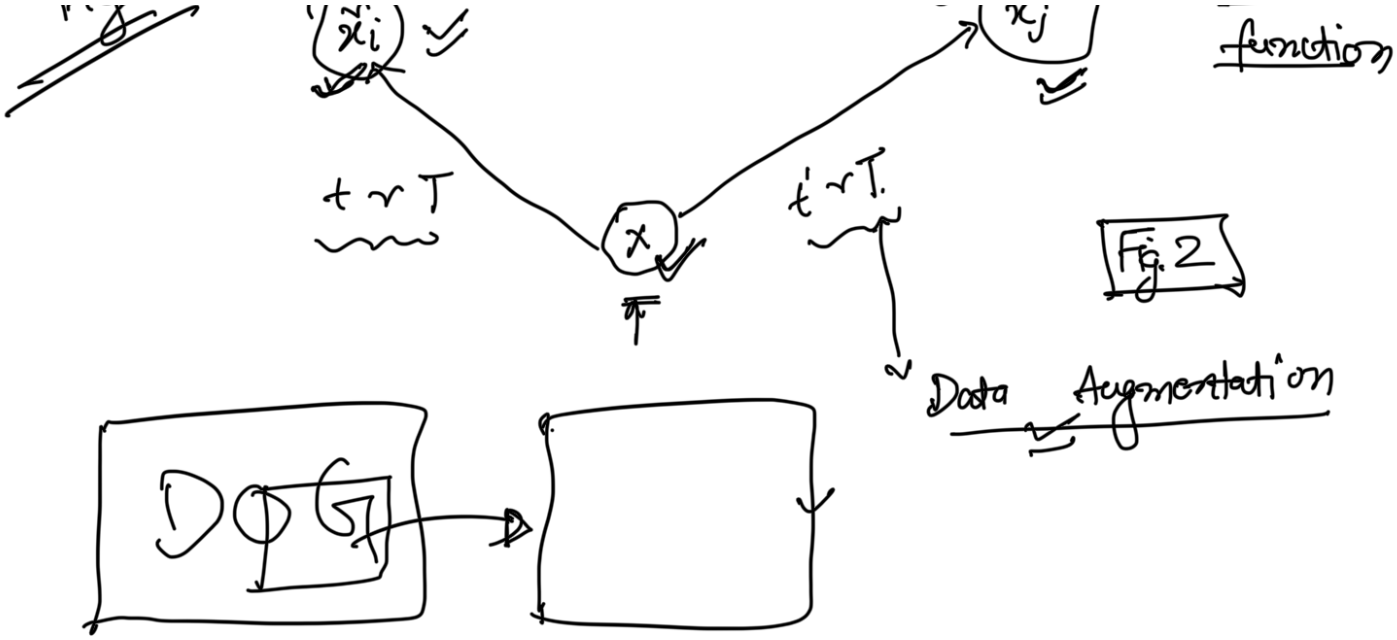
2. Learning a **non-linear transformation** :

3. Batch size ↑ , model performance increases

Image Net]



Fig



Data Augmentations

1. radom crop followed by resizing
2. random color distortions
3. random Gaussian blur

$$z_i = g(h_i) = W^{(2)} \sigma(W^{(1)} h_i)$$

$\sigma \rightarrow$ RELU

Contrastive loss function →

minibatch of N → $2N$ data points

Every sample: 2 data points positive sample

$2N-2$ Negative samples

$$l_{i,j} = -\log \frac{\exp(\mathrm{Sim}(z_i, z_j)/\tau)}{2N \ldots}$$

$$\sum_{k=1} \mathbb{1}_{[k \neq i]} \exp\left(\mathrm{sim}(z_i, z_k)/\tau\right)$$

$$\mathrm{sim}(u,v) = \frac{u^T v}{\|u\| \, \|v\|}$$

Indicator function

$\mathbb{1}_{k \neq i} \in \{0, 1\}$ 1 $k \neq i$

$\tau \rightarrow$ temperature parameter

NT-Xent (Normalized temperature - Scaled cross entropy loss)

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for every minibatch (N)

for all $k \in \{1, \ldots N\}$

$t \sim T, t' \sim T$

$$\begin{cases} \tilde{x}_{2k-1} = t(x_k) \\ h_{2k-1} = f(\tilde{x}_{2k-1}) \\ z_{2k-1} = g(h_{2k-1}) \end{cases}$$

$$\begin{cases} \tilde{x}_{2k} = t'(x_k) \\ h_{2k} = f(\tilde{x}_{2k}) \\ z_{2k} = g(h_{2k}) \end{cases}$$

end for

L (loss)

update loss

return encoder network] $g(\cdot)$]

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* Training with larger batch sizes (32, 64, 128)

256 → 8192

* Randomly crop + resize in one branch	Same image

Table		Representation
prediction	Random	h $g(h)$
color vs grayscale		